Christopher Fontana

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Education

Georgia Institute of Technology **Expected May 2029**
B.S. Aerospace Engineering, B.S. Physics Atlanta, GA
Relevant Coursework: Statistical Mechanics, Differential Equations, Classical Mechanics, Electrostatics

Clubs & Activities

GT Supersonics – Software. Writing C++ for an ArduPilot flight-control rewrite for aircraft approaching supersonic speed. Helping Ground, Controls, and Navigation turn their protocols into code on that stack.

RoboJackets RoboCup – Firmware. Writing C++ drivers for the club's new electrical stack, and moving the older Rust firmware into that C++ codebase.

RoboJackets Outreach – Volunteer at FTC Robot in 30 Hours, where I helped about 50 people with robot design. Working on silicone molding to cut part cost for FRC Robot in 30 Hours.

Projects

UAV Wing Aerodynamic Analysis | SOLIDWORKS, ANSYS Fluent, OpenFOAM **Aug 2026 – Present**
- Designed a UAV wing in SOLIDWORKS and ran the same geometry in ANSYS Fluent and OpenFOAM across angles of attack and freestream speeds.
- Compared CL, CD, stall, and maximum L/D between the two solvers.

Iron | Rust **2024 – 2026**
- Wrote a Rust interpreter with a lexer, parser, evaluator, classes, and method dispatch.
- Built a standard library that runs programs with objects and I/O.

Apex Pathing | TypeScript, Next.js, Java, Kotlin **March–August 2026**
- Led the dev team on an open-source FTC path-planning and kinematics library and reviewed the changes they landed in the SDK.

Familyr (HackGT 13) | Next.js, Supabase, Postgres, Google Calendar API **Sep 2025**
- Built the Next.js frontend and the Supabase/Postgres backend: schema, APIs, chat, and two-way Google Calendar sync.
- Events pulled from chat require consent and opt-out. Private messages stay out of the health reports.

Experience

Makerspace Technical Assistant **Aug 2025 – May 2026**
Florida Gulf Coast University Ft. Myers, FL
- Operated Bambu Lab X1C and Prusa MK3S+ printers, a laser cutter, a vinyl cutter, and a desktop CNC mill for student and research jobs.
- Calibrated those machines, did preventive maintenance between jobs, and certified 10+ students a semester on the laser cutter.

STEM Camp Counselor **Summers 2022 – 2026**
Edison and Ford Winter Estates Ft. Myers, FL
- Ran CAD, FDM, and microcontroller build challenges, and helped students debug failed prints, wiring, and joints.

Technical Skills

Programming: C++, Rust, TypeScript, Java, Kotlin
CAD: SOLIDWORKS
Manufacturing: FDM Printing, Laser Cutting, CNC Milling
Simulation: ANSYS Fluent, OpenFOAM
Tools: Next.js, Supabase, Postgres, ArduPilot